

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

103000903 - Graph Analysis And Social Networks

DEGREE PROGRAMME

10BA - Master Universitario En Ciencia De Datos

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 2

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1. Description

1.1. Subject details

Name of the subject	103000903 - Graph Analysis And Social Networks
No of credits	3 ECTS
Type	Optional/elective
Academic year of the programme	First year
Semester of tuition	Semester 2
Tuition period	February-June
Tuition languages	English
Degree programme	10BA - Master Universitario en Ciencia de Datos
Centre	10 - E.T.S. De Ingenieros Informáticos
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Emilio Serrano Fernandez		emilio.serrano@upm.es	Sin horario.
Javier Bajo Perez (Subject coordinator)		javier.bajo@upm.es	- -
Laura Melgar Garcia		laura.melgar@upm.es	Sin horario.

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Skills and learning outcomes *

3.1. Skills to be learned

CECD04 - Capacidad para aplicar técnicas para la generación de visualizaciones adecuadas a cada problema para el análisis y la exploración de datos, y para la correcta comunicación de los resultados del análisis.

CG07 - Aplicación de los últimos o más novedosos métodos para resolver problemas que, posiblemente, involucren a otras disciplinas

CG11 - Conocimiento y comprensión de la informática para crear modelos, así como sistemas y procesos de información complejos

3.2. Learning outcomes

RA10 - Ser capaz de establecer un debate fundamentado sobre el conocimiento científico y las bases de la investigación

RA34 - Apply AI techniques in real world data scenarios

RA21 - Conocer cómo se aplican las técnicas de computación científica en algún campo específico de ciencia o ingeniería

RA17 - Conocer los fundamentos de las técnicas de visualización analítica

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

4. Brief description of the subject and syllabus

4.1. Brief description of the subject

Social computing is a general term for an area of computer science that is concerned with the intersection of social behavior and computational systems. During recent years the Internet introduced a social element where users could network, share interests, publish personal insights and use their computers for more than just doing a job faster, and this has led to the development of social machines where both humans and machines collaborate to solve social problems. This course presents the principals of social computing and focuses on graph and network analysis.

4.2. Syllabus

1. Introduction to Social Computing.
2. Graph mining and Social Network Analysis.
3. Sentiment Analysis in Social Networks.
4. Trust and reputation in social networks
5. Practical assignment

5. Schedule

5.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Introduction to social computing. Duration: 03:00 Lecture			
2	Graph mining and Social Network Analysis Duration: 03:00 Problem-solving class			
3	Graph mining and Social Network Analysis Duration: 03:00 Lecture			
4	Research Work Duration: 04:00 Problem-solving class			Evaluation in class. Research Work. Individual presentation Progressive assessment Presential Duration: 03:00
5	Sentiment Analysis in Social Networks. Duration: 03:00 Lecture			
6	Trust and reputation in social networks. Duration: 03:00 Lecture			
7	Practical Assignment Duration: 03:00 Cooperative activities			
8	Practical Work Duration: 03:00 Problem-solving class			Evaluation in class. Test exam. Written test Progressive assessment Presential Duration: 03:00
9				
10				
11				
12				
13				
14				
15				
16				
17	Practical Work Duration: 03:00 Problem-solving class			Research Work Individual presentation Global examination Presential Duration: 03:00 Test exam Written test Global examination

				Presential Duration: 03:00
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Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

6. Activities and assessment criteria

6.1. Assessment activities

6.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
4	Evaluation in class. Research Work.	Individual presentation	Face-to-face	03:00	50%	5 / 10	CG07 CG11 CECD04
8	Evaluation in class. Test exam.	Written test	Face-to-face	03:00	50%	5 / 10	CG07 CG11 CECD04

6.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Research Work	Individual presentation	Face-to-face	03:00	50%	5 / 10	CG07 CG11 CECD04
17	Test exam	Written test	Face-to-face	03:00	50%	5 / 10	CG07 CG11 CECD04

6.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Research Work	Individual presentation	Face-to-face	03:00	50%	5 / 10	CG07 CG11 CECD04
Practical Work	Group presentation	Face-to-face	03:00	50%	5 / 10	CG07 CG11 CECD04

6.2. Assessment criteria

Continuous Evaluacion

Research work (Exam):

Individual research work about one of the topics presented in class

The students can choose a topic of interest for them.

50% of the grade of the subject.

Test Exam

Test about graph minning and/or analysis of social networks.

50% of the grade of the subject

Recovey exam

The student has a second chance to present one of the works (Research work or Practical work).

Extraordinary Exams.

Individual research work (50%)

Test Exam (50%)

7. Teaching resources

7.1. Teaching resources for the subject

Name	Type	Notes
Slides	Web resource	